

University of Niagara Falls Canada
Master of Data Analytics
CPSC 510 -3 Data Warehousing & Visualization
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Assignment 3: Credit Risk Visualization and Advanced Analysis

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1. DATA DESCRIPTION

The dataset is made up of individual credit card customer records from a financial institution. The record includes credit limits, repayment histories, demographics and payment default status. Data cleaning involved handling of nulls, encoding categorical data, renaming columns names and transforming variables (See Table 1, Appendix A)

2. OUTLIER DETECTION

Outliers were detected in the credit card limits from the visuals of the box plot below but were kept for analysis because they allowed for segregation of the data. Empty columns, nulls were removed, and appropriate data types were used. Also, columns were renamed as stated in the table above for better understanding.

3. CALCULATED COLUMNS DAX MEASURES

We introduced twelve (12) calculated columns like Risk_Cluster, Avg_Replacement_Status, Avg_Bill, Avg_Payment, Avg_Utilization, Payment_Ratio etc and Eleven(11) measures DefaultRate, DefaultRate_PerCluster(%), TotalBilled, TotalPaid, TotalDefaults etc just to mention a few.

4. CUSTOMER RISK PROFILING

The dashboard reveals key insights into customer credit behavior, including average credit limits, payment trends, and default risk distribution (See Fig.1, Appendix B), with younger customers tending to utilize credit more aggressively and high-risk clusters requiring targeted risk management, while most customers are low-risk and payment behavior varies despite high credit limits.

5. TREND ANALYSIS

The dashboard reveals that while the majority of customers repay their credit, with a total credit limit of \$4 billion for non-defaulters, a significant amount of \$863 million was extended to customers who defaulted, highlighting the need for monitoring, particularly in very low and very high credit segments, and suggesting better screening or limits for certain customer segments to mitigate the 17% default rate (See Fig.2, Appendix B)

6. CLUSTERING OF CUSTOMERS

This Power BI dashboard tab, titled “PCA Clustering”, visualizes the result of PCA (Principal Component Analysis) and Clustering to segment customers based on their credit behavior. The X-axis (PC1) and Y-axis (PC2) represent the first two principal components from a principal component analysis (PCA) transformation. Each dot represents a customer in the transformed feature space. The colors (Cluster 0, 1, 2) indicate cluster groupings generated from K-means using the Python code below;

```
from sklearn.preprocessing import StandardScaler
from sklearn.decomposition import PCA
from sklearn.cluster import KMeans

# dataset = your input table in Power BI
features = dataset[['Avg_Payment_Ratio', 'Avg_Credit_Limit', 'Default_Rate', 'AGE', 'CREDIT_LIMIT']]

# Scale
scaler = StandardScaler()
X_scaled = scaler.fit_transform(features)

# PCA
pc1 = PCA(n_components=2)
pc1_result = pc1.fit_transform(X_scaled)
dataset['PC1'] = pc1_result[:, 0]
dataset['PC2'] = pc1_result[:, 1]

# K-Means
kmeans = KMeans(n_clusters=3, random_stat
```

Customers are grouped into 3 distinct behavioral clusters based on credit-related attributes.

From the visual (See Fig.3, Appendix A), it was observed that;

Clusters differ in spatial grouping, suggesting differences in financial behavior and default payment profiles.

- **Cluster 0** is dense and concentrated near the origin (PC1 ~ 0 to 10, PC2 ~ -5 to 5), suggesting customers here might have typical or average behaviors. A few outliers exist toward higher PC1 values, hinting at slight variance in behavior within the cluster.

Interpretation: Represents **average or typical customers** with moderate variability of some customers slightly diverge in behaviors (e.g., higher spending or slightly more credit usage). This is likely the "middle ground" cluster.

- **Cluster 1** is tightly grouped near the origin (PC1 ~ -5 to 10, PC2 ~ -10 to 5), with a compact and dense cluster extending from a dense bottom-left corner, with a triangular or fan-like spread indicating low variation in customer behavior within this group

- **Interpretation:** Represents a **very** homogeneous group of customers with similar behavior. The spread from the corner could indicate a core pattern with mild deviations, such as a gradual increase in payment or balance habits. Possibly low-risk or low-usage customers (given lower PC1/PC2 values).

- **Cluster 2** is densely population at low PC1/PC2 values—possibly more conservative or low-risk customers with a small number of outliers suggesting diversity within the cluster. Total credit exposure of \$2bn indicates a high financial exposure.

- **Interpretation:** Likely includes high-value or high-credit-limit customers. Tight cluster indicates shared behavior (e.g., consistent spending or repayment patterns). Outliers suggest a few risky or high-variance individuals within an otherwise stable group.

Feature	Cluster 0	Cluster 1	Cluster 2
Spread	Moderate	Very Tight	Tight with some outliers
Behavior Type	Average/typical	Homogenous/stable	High-value, mostly stable
Risk Level	Low-to-moderate	Likely low	Mixed: mostly low, few high
Variance	Slight variance	Very low	Low with high-end outliers
Strategic Role	Baseline customer	Low-risk customers	Premium segment, monitor outliers

7. AI-DRIVEN INSIGHTS AND INSIGHT ENHANCEMENT

Cluster 0: The Baseline Segment

Policy: This group can be used as a benchmark group to maintain standard credit policies and assess risks in other segments.

Actionable Insights: Monitor outliers for potential risk or opportunity (e.g., offer higher credit to those showing good trends) and offer generic loyalty or cashback incentives to customers. Consider campaigns like “You're pre-qualified!” under this category.

Cluster 1: The Conservative Segment

Policy: This group can be used for pre-approving credit increases because of their low variability and consistent behavior reduces risk with minimal monitoring needs.

Actionable Insights: Offers financial products with favorable interest rates, appealing to debt-conscious customers. We can leverage these relationships to promote savings, mortgages, or investment opportunities. Consider campaigns like “Unlock VIP perks” under this category.

Cluster 2: The High-Value Segment

Policy: Here, we need to balance rewarding the majority of reliable customers with vigilant oversight of outlier accounts that may pose risks.

Action: Offer high-value perks like premium cards and personalized concierge services to valued customers. Additionally, consider tailored credit limits based on customer performance or identify and address potential risk patterns proactively. Consider campaigns like “Boost your rewards” under this category.

8. Business Impact

Using this data driven segmentation, companies can be proactive in implementing different approaches that enhance customer satisfaction and protect against financial loss. The framework supports the following.

- i. Targeted credit policy deployment based on behavioral risk
- ii. Efficient allocation of marketing and loyalty budgets
- iii. Early intervention mechanisms for potentially high-risk accounts

Appendix A

Table 1: Data Description and Transformation

Provided Dataset Column Name	New Column Name	Description
ID	ID	Unique Client ID
Limit_Bal	Credit_Limit	Amount available to each client as credit
Sex	Sex	Gender (1=male, 2=female, 3=others)
Education	Education	Education level (1 = graduate school, 2 = university, 3 = high school, 4 = others)
Marriage	Marriage	Marital status (1 = married, 2 = single, 3 = others)
Age	Age	Age in years
Pay_0 to Pay_6	Bill_Amt_Apr to Bill_Amt_Sept	The repayment status of each client from month 1 to month 6.
Bill_Amt1 to Bill_Amt6	Pay_Amt_Apr to Pay_Amt_Sept	Amount of the bill statement (past 6 months)
Pay_Amt1 to Pay_Amt6	Repayment_Amt_Apr to Repayment_Amt_Sept	Amount paid in the previous 6 months
Default Payment Next Month	Default Payment Next Month	Target variable: whether the client defaulted (1) or not (0)

Appendix B

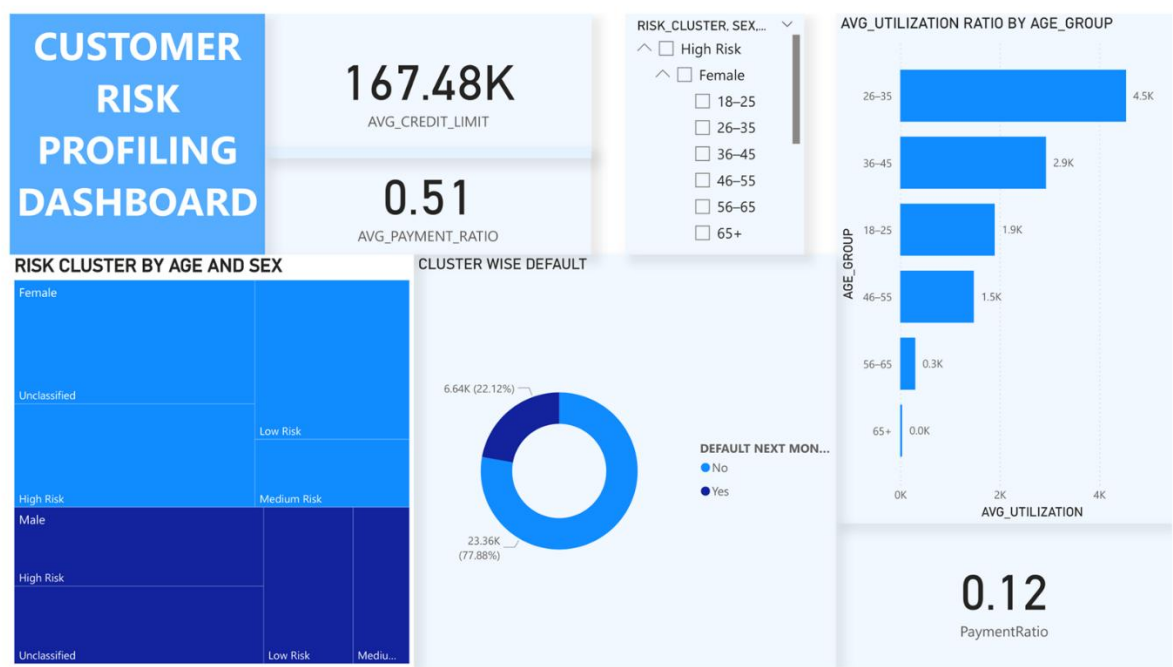


Figure 1: Customer Risk Profiling

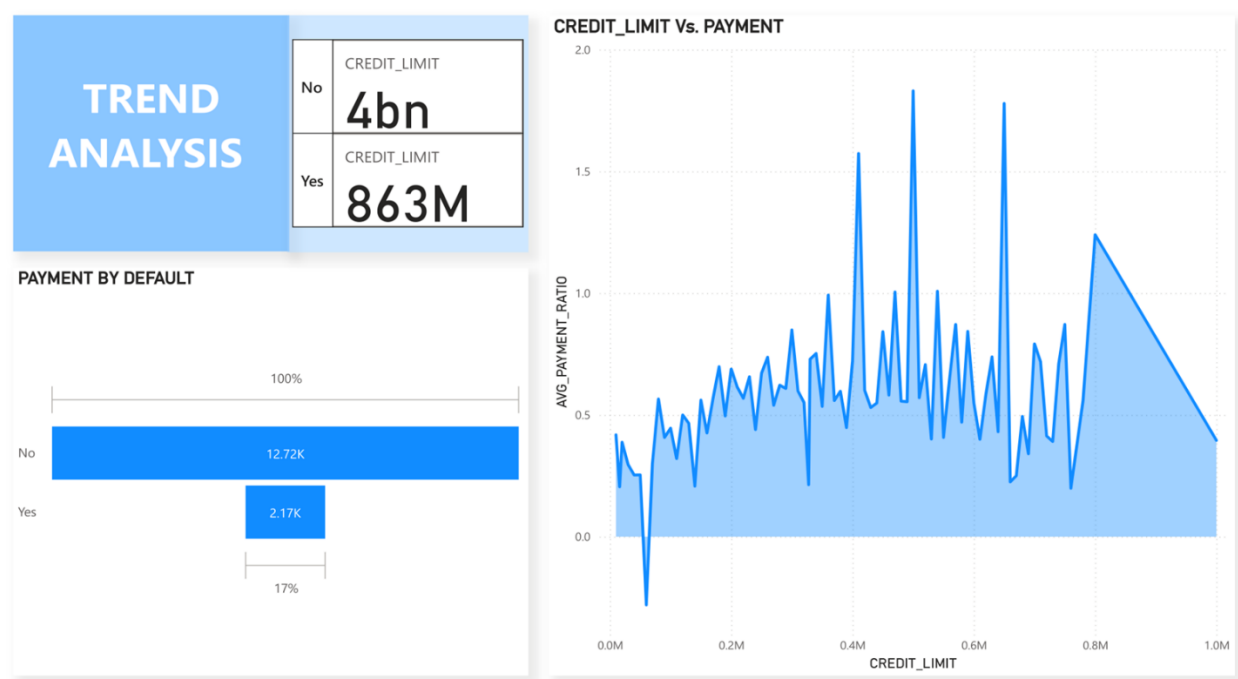


Figure 2: Customers Repayment History Showing Payment By Default

Appendix C

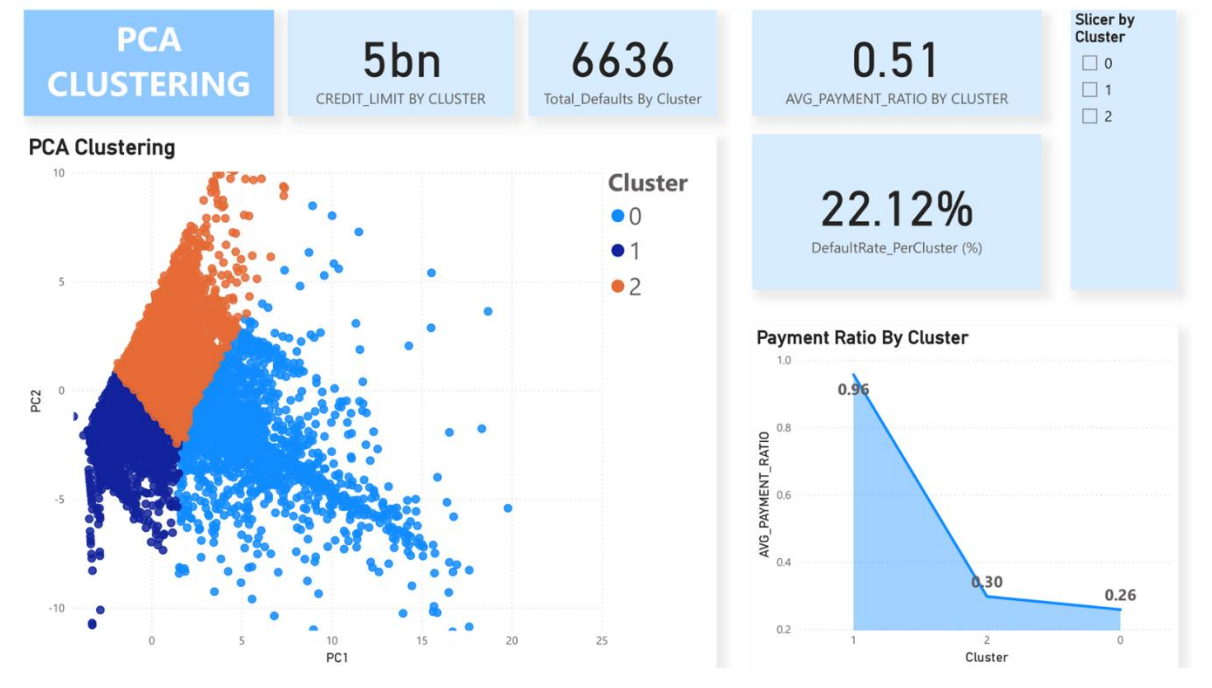


Figure 3: PCA Scatterplot Showing K-Means Clusters Based on Credit Behavior